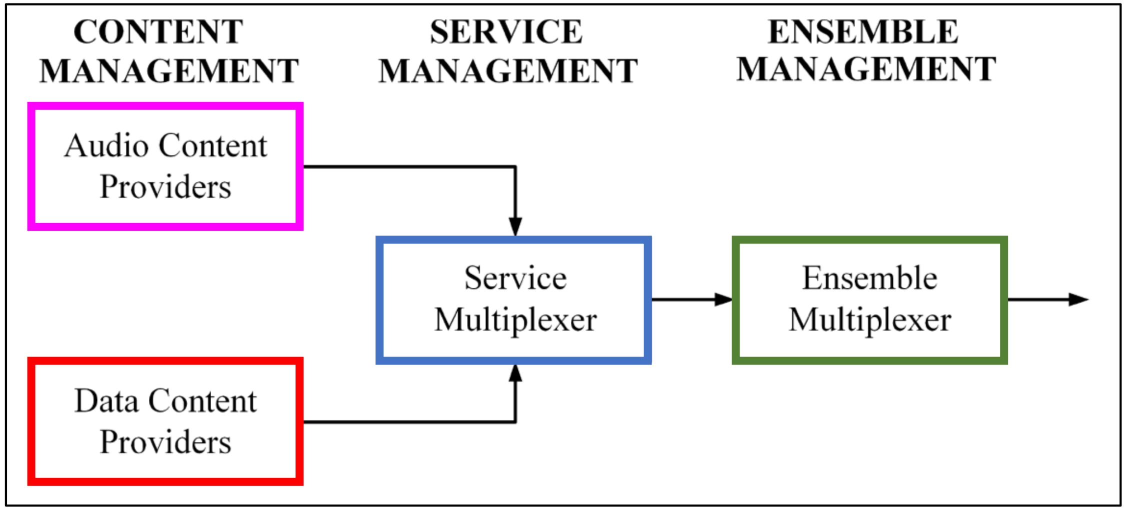
Fast Information Channel (FIC):

- Rapid access of information by a receiver (used to send the Multiplex Configuration Information (MCI) and Service Information)
- Transmitted at the beginning of each DAB frame using robust Forward Error Correction **(FEC)** parameters.

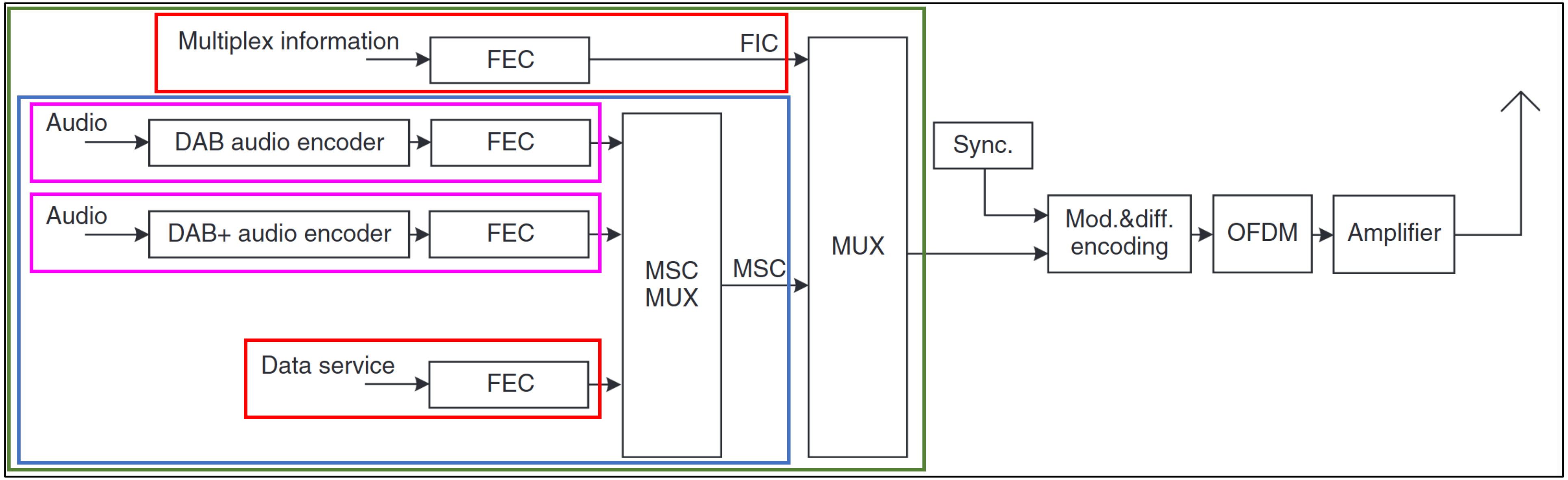


Main Service Channel (MSC):

- Carries the payload data (audio and data service components).
- Divided into a number of subchannels that contain the different services.
- Each sub-channel is then individually FEC encoded and has its own time interleaver,

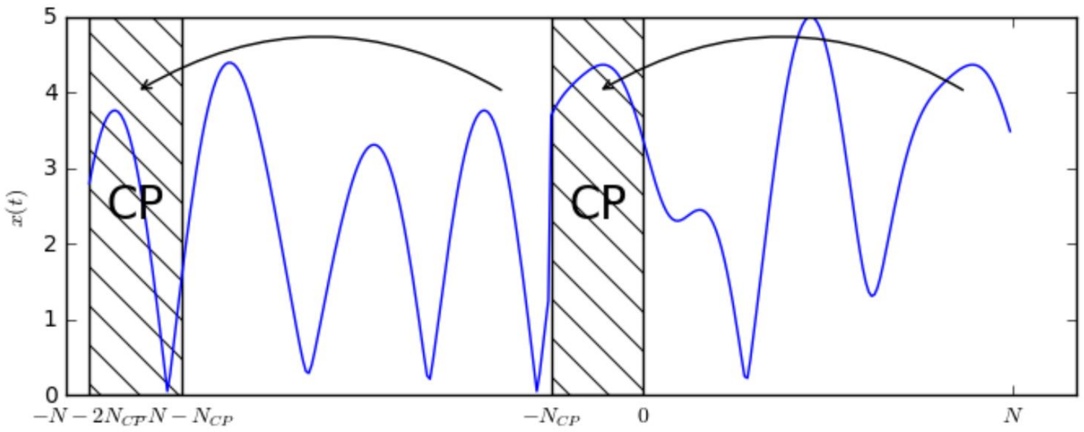
Error protection: The binary stream is protected by :

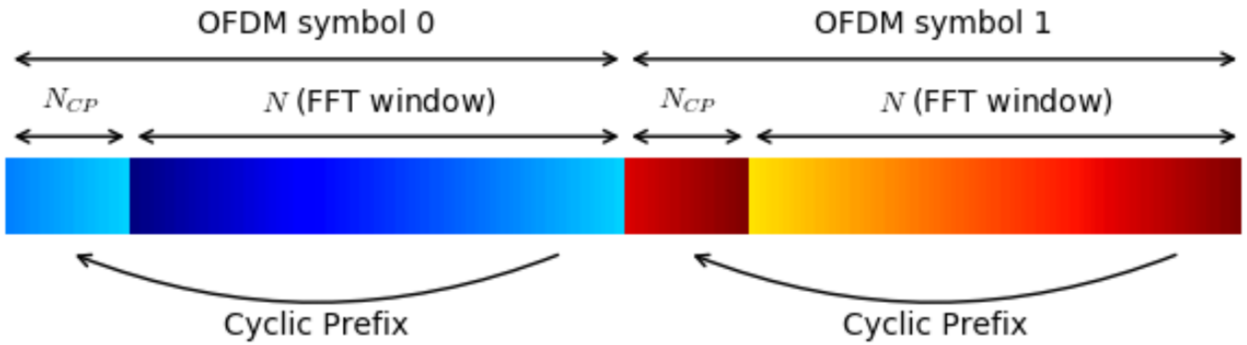
- Convolutional coding
- Additional Reed Solomon
- Time interleaving

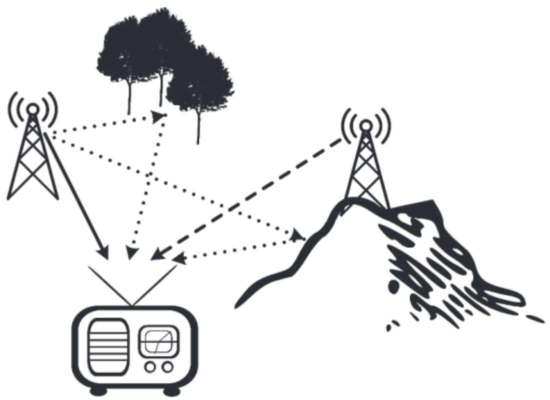


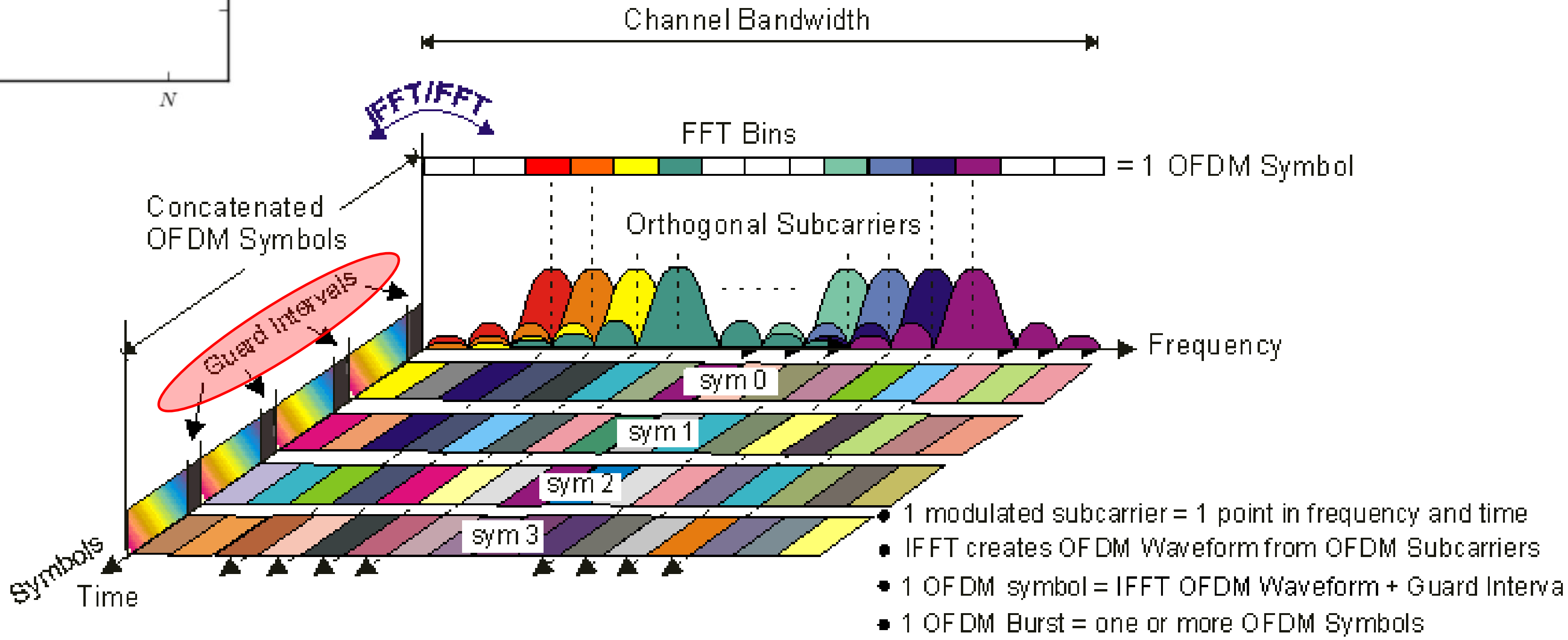
DAB and OFDM:

- Multipath issue -> **ISI (Inter Symbol Interference) ->** **Guard interval**
- Multiple transmitters radiate identical signals on the same frequency -> **Using SFN (Single Frequency Networks)**
  - Each path in an SFN causes only a simple phase shift per subcarrier.
  - The induced phase shift varies from one subcarrier to another, but not within a subcarrier.
  - With the presence of the **CP (Cyclic Prefix)** handling the time spread, there is no spectral overlap : turns the linear convolution with the channel into a circular convolution

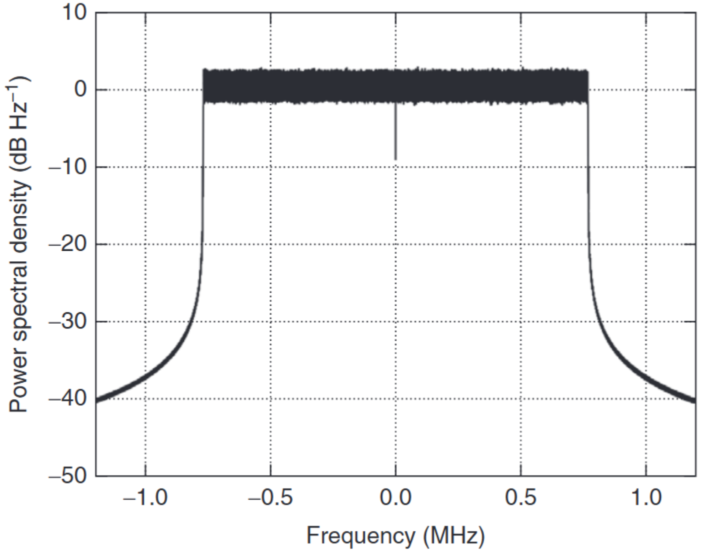






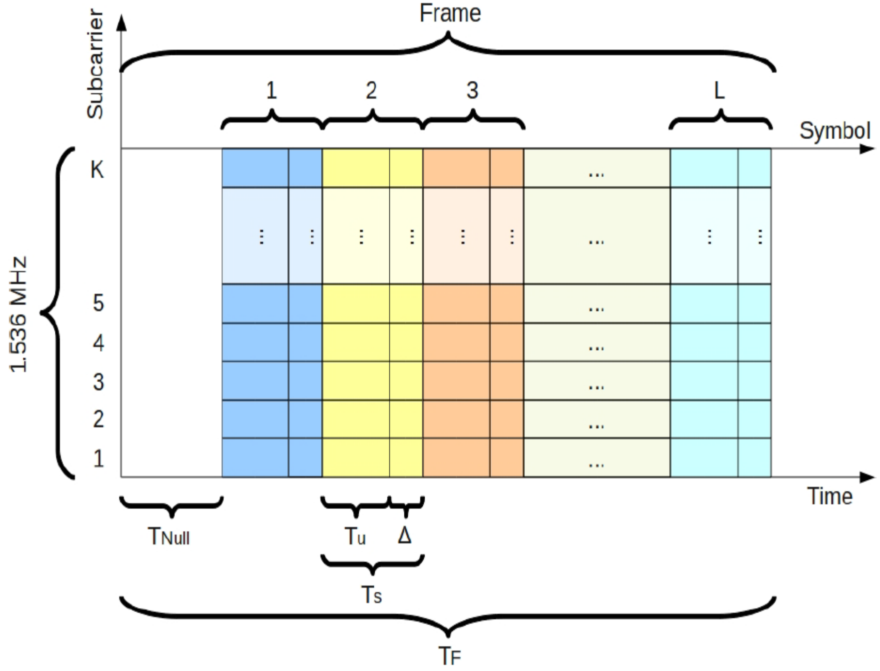


```
Tf    = 196608/Fs;  % transmission frame duration
Tnull = 2656/Fs;    % null symbol duration
Tu    = 2048/Fs;    % symbol duration excluding ICI (Intercarrier Interference)
Tg    = 504/Fs;     % duration of ICI
Tsym  = 2552/Fs;    % duration (Tu+Tg) of OFDM symbols of indices l = 1, 2, 3, …, L
Fu    = Fs/2048;    % Intercarrier spacing 1/Tu
Fs    = 2048000;    % sampling rate of IQ signal
K     = 1536;       % number of transmitted carriers
L     = 76;         % number of OFDM symbols per transmission frame (the Null
                      symbol being excluded);
```



Encoding:

- Each active **subcarrier** carries one complex symbol from differential **QPSK**
- **1536 complex symbols** as spectral data
- **Frequency Interleaver** : prevents localized frequency interference from causing large contiguous data losses by spreading errors across different subcarriers.
- **Mapping** to frequency vector of **2048 samples** :
  256 zeros - 768 data - 1 zero (DC) - 768 data - 255 zeros
- **Operate IFFT** -> time domain to what supposed to be harmonic info.
- **With CP :** vector of 2552 with 504 last values of time before all time data
- **First symbol** used as phase reference of differential-QPSK (only 75 symbols are data)



$$y_{l,k} = q_{l,n} \text{ for } l = 2,3,4,\cdots,L$$

$$y_{l,k} = q_{l,n} \text{ for } l = 2,3,4,\cdots,L$$

$$n \in \{0,1,2,\ldots,1535\}$$

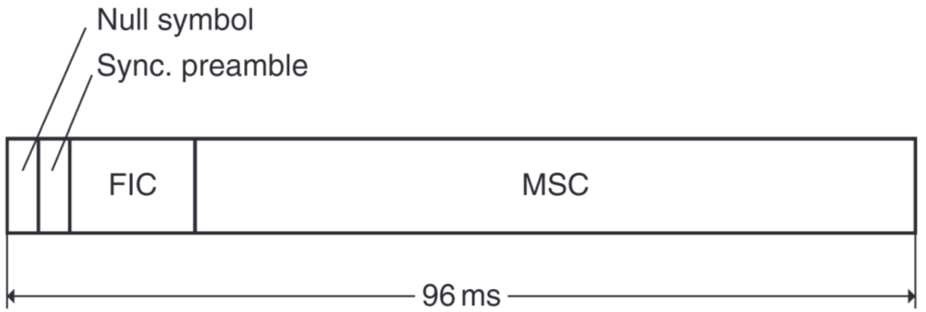
$$k \in \{-768, -767, -766, \ldots, 768\} \setminus \{0\},$$

Table 25: Frequency interleaving for transmission mode I

i	Π(i)	d_n	n	k
0	0			
1	511	511	0	-513
2	1 010	1 010	1	-14
3	1 353	1 353	2	329
4	1 716	1 716	3	692
5	291	291	4	-733
6	198			
7	1 037	1 037	5	13
8	1 704	1 704	6	680
9	135			
10	218			
11	1 297	1 297	7	273
12	988	988	8	-36
13	1 076	1 067	9	43
14	46			
15	1 109	1 109	10	85
16	592	592	11	-432
17	15			
18	706	706	12	-318
:	:	:		
2044	1 676	1 676	1 533	652
2045	1 819			
2046	1 630	1 630	1 534	606
2047	1 221	1 221	1 535	197

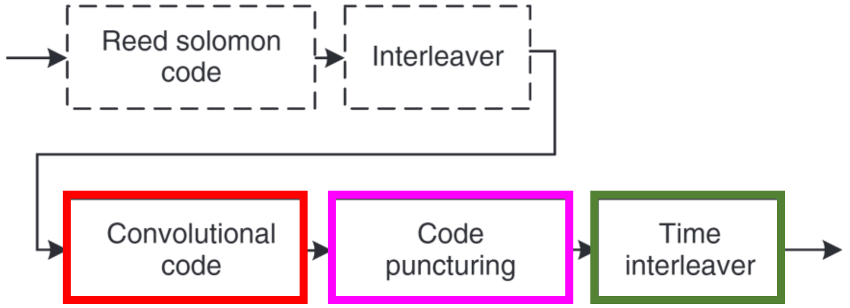
Synchronization: NULL symbol(1,297 ms) + Synchronization Preamble

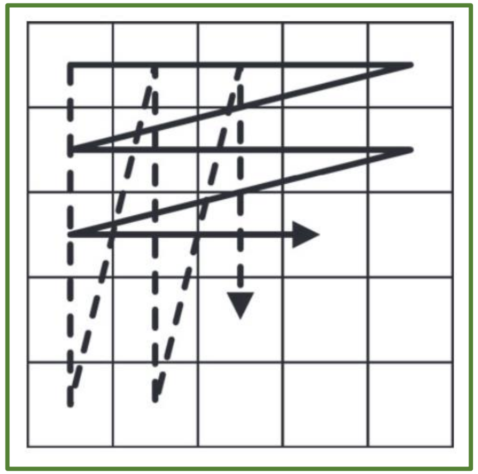
- Zero power detection
- FIC / MSC: Different SP (also used as reference of D-QPSK)

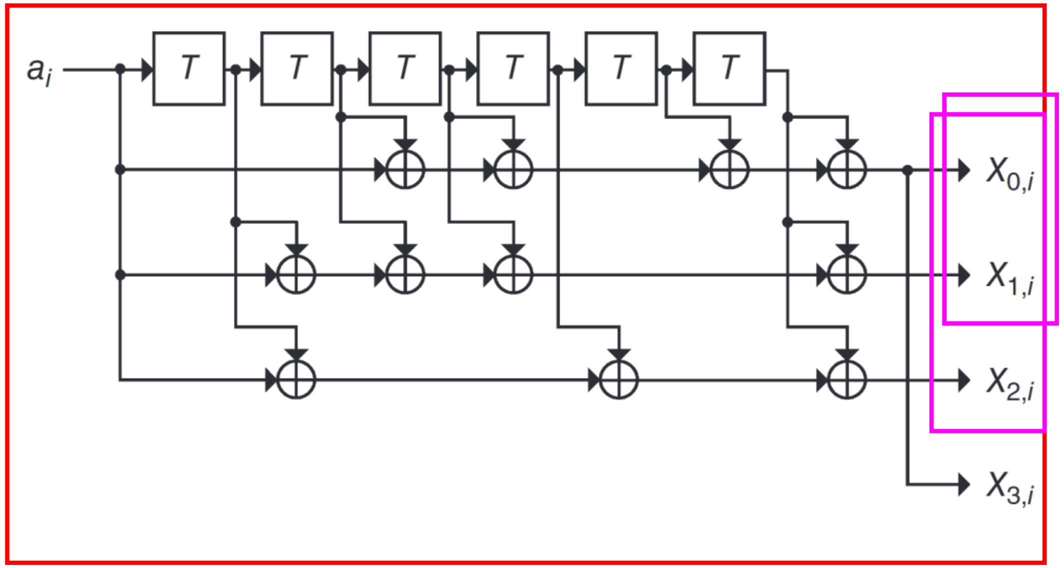


Forward Error Correction (FEC): Encoding MSC and FIC

- Convolutional code
- Code puncturing
- Time interleave (only for MSC not for FIC) : adjacent error bits issue
  The data is written in rows(solid lines) and read out in columns (dashed lines).
- Reed Solomon code & interleaver (DAB+)



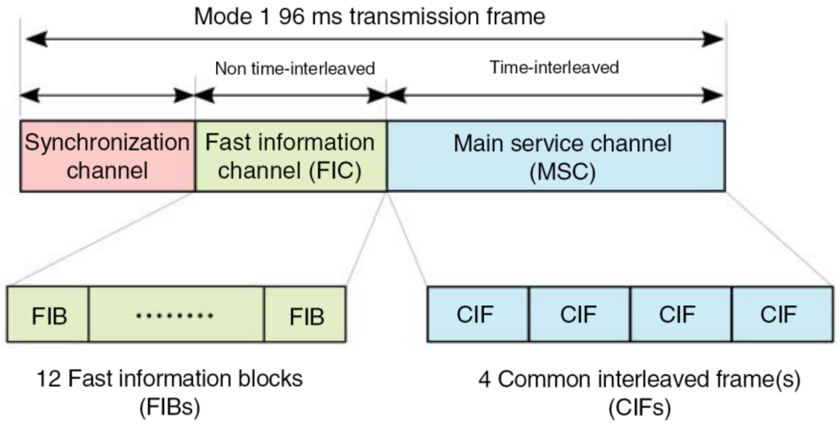


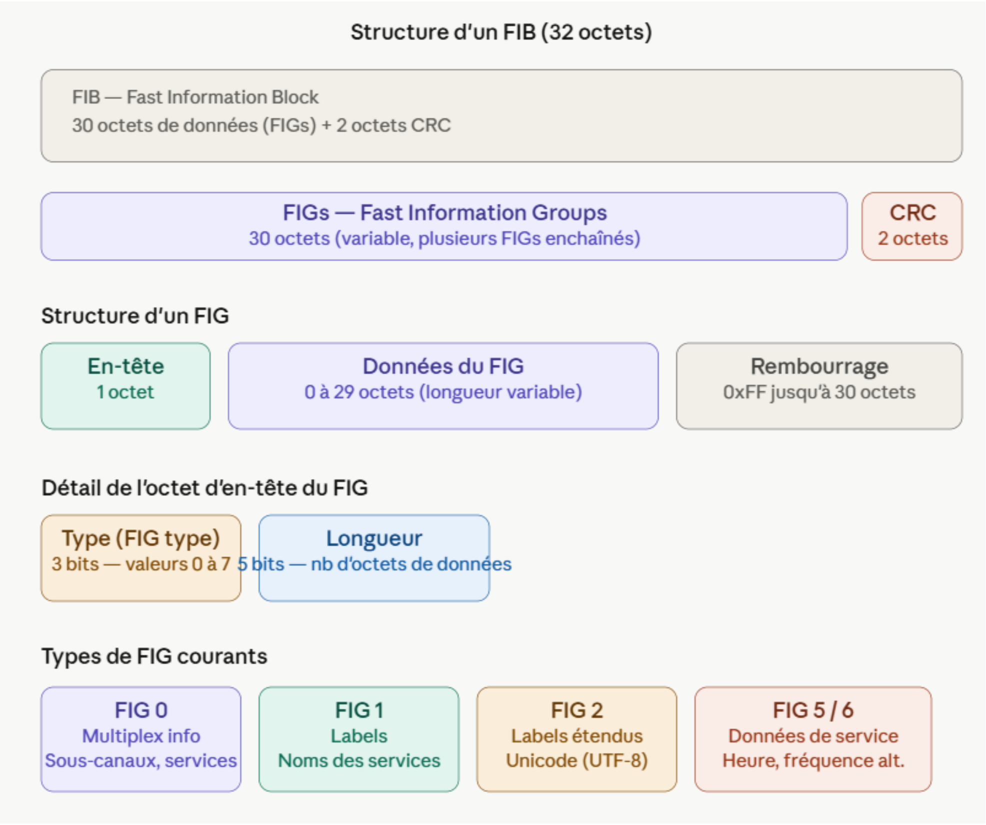


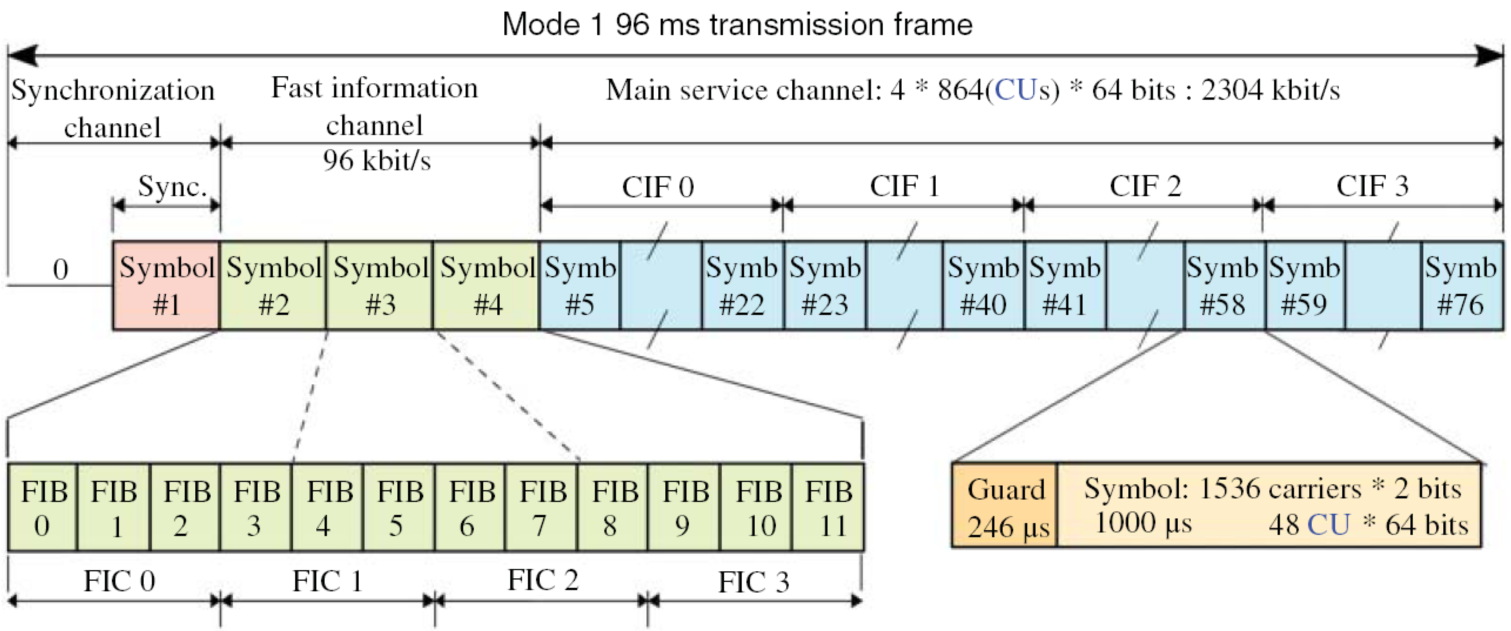
LAB : may 4th

- Compléter les deux fonctions "SymbolFFT.m" et "myFreqDeInterleave.m" du code fournis pour extraire les données du FIC
- Documenter, expliquer et justifier chacune des étapes

```
CRC OK
FrameNr:  2 FIB:  1 Type:  0 Extension:  0 C/N: 0 OE: 0 P/D: 0 EId: 52757 ChangeFlag: 0 ALFlag: 0 CIFCount:  1323 OccurenceChange: 189
FrameNr:  2 FIB:  1 Type:  0 Extension:  2 C/N: 0 OE: 0 P/D: 0 SId: 49701 MSC stream audio ASCTy:  0 SubChId:  5 P/S: 1 CAFlag: 0
                                                               SId: 49701 MSC stream audio ASCTy:  0 SubChId:  8 P/S: 0 CAFlag: 0
                                                               SId: 49720 MSC stream audio ASCTy:  0 SubChId:  9 P/S: 1 CAFlag: 0
                                                               SId: 49704 MSC stream audio ASCTy:  0 SubChId:  8 P/S: 1 CAFlag: 0
FrameNr:  2 FIB:  1 Type:  0 Extension:  1 C/N: 0 OE: 0 P/D: 0 SubChID:  1 StartAdr:   0 UEP ProtLevel: 3   SubChSize:  96 BitRate: 128
FrameNr:  2 FIB:  4 Type:  0 Extension:  1 C/N: 0 OE: 0 P/D: 0 SubChID:  2 StartAdr:  96 UEP ProtLevel: 3   SubChSize:  96 BitRate: 128
                                                               SubChID:  3 StartAdr: 192 UEP ProtLevel: 3   SubChSize: 116 BitRate: 160
                                                               SubChID: 62 StartAdr: 846 EEP ProtLevel: 3-A SubChSize:  12
                                                               SubChID:  4 StartAdr: 308 UEP ProtLevel: 3   SubChSize:  96 BitRate: 128
                                                               SubChID:  5 StartAdr: 404 UEP ProtLevel: 3   SubChSize:  48 BitRate:  64
                                                               SubChID:  8 StartAdr: 500 UEP ProtLevel: 3   SubChSize:  48 BitRate:  64
                                                               SubChID: 11 StartAdr: 692 UEP ProtLevel: 3   SubChSize:  96 BitRate: 128
                                                               SubChID: 12 StartAdr: 788 UEP ProtLevel: 3   SubChSize:  58 BitRate:  80
```







Mode I with $T = 1/2\,048\,000$ s.

System parameter	Mode I
Frame duration T_F	196 608 T
	96 ms
OFDM symbols per frame L	76
Number of OFDM subcarriers K	1536
Null symbol duration T_{NULL}	2656 T
	~ 1297 µs
Guard interval duration T_G	504 T
	246 µs
Useful symbol duration T_U	2048 T
	1 ms
Total symbol duration T_S	2552 T
	~ 1246 µs

LAB : may 11th

- Implémenter un model de canal avec
  1. une dérive sub-symbole de la fréquence de l'oscillateur local
  2. une dérive multi-symbole de la fréquence de l'oscillateur local sachant que la documentation technique de l'oscillateur local indique une dérive de 50ppm à 230MHz
  3. un bruit blanc gaussien additif (AWGN, Additive White Gaussian Noise)
- Evaluer les limites de décodage de la trame IQ de 16s fournie -sans dérive de fréquence et sans bruit- (si besoin utiliser le package *mmbtools* pour générer votre propre trame)
- Implémenter une correction fine de la dérive de fréquence (Fine Frequency Synchronization) et tester sur tout l'intervalle de dérive sub-symbole. Etablir la caractéristique de la limite de synchronisation (intervalle de fréquence) en fonction du niveau de bruit.
- Implémenter une correction grossière de la dérive de fréquence (Coarse Frequency Offset) et faire le même test (sans dérive sub-symbole)

Fréquence d'échantillonnage Fs	2,048 MHz
Nombre de sous-porteuses actives	1536
Taille d'un symbole (Nu)	2048
Espacement inter-porteuse Δf_{SYM}	Fs / Nu = 1000 Hz
Durée symbole utile Tu	$1 / \Delta f_{SYM}$ = 1 ms
Préfixe cyclique Tg	246 µs
Durée symbole total Ts	1,246 ms

LAB : may 18th

Egalisation du canal de transmission

1. caractéristique SER en fonction de la vitesse du véhicule
2. idem en fonction des délais multi trajet
3. superposer dans ces deux figures les courbes pour 4 fois moins de pilotes et pour 4 fois plus
4. Refixer la concentration des pilotes et refaire les courbes q1 et q2 en faisant une estimation de H au bout de m=6 transmissions avec un SNR de 25dB
5. refaire 4 avec cette fois des positions des pilotes qui changent m fois (configurations distinctes à chaque fois) avec le même SNR